

Course Code : ECT 451-1

KQLR/RS – 24 / 1672

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

EMBEDDED SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) Assume suitable data wherever necessary.

1. Retrieve the ARM features along with ARM Family Comparison. 6(CO1)
2. Clarify the difference between Pointers and Arrays in embedded C language. 6(CO3)
3. Interpret what do you understand by volatile variable ? Explain where and why it is used with the help of an example. 6(CO2)
4. Using Tail chaining explain with neat waveform how nested vector interrupts execution of external interrupts take place. 6(CO2)
5. Detect how floating point to fixed point conversion is carried out considering Q4.3 and solve for value -3.97 . 6(CO3)
6. Assume a PWM pin of Cortex M your choice and generate a waveform of 1 KHz by using an embedded C code. Explain the mode you will use to achieve the above PWM signal with the help of neat waveform. 10(CO3,4)

7. Employ with an embedded C program for led blinking with the help of an external interrupt and explain the working in regards to ISR. 10(CO3,4)
8. Design a circuit such that a switch is connected to a digital input pin of Cortex M and an LCD is connected on a GPIO port. Write an embedded code that displays the updating count of the switch pressed on LCD. 10(CO4,5)

